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Project: 226052 DFCI-BIDMC New Cancer Hospital - 1 Joslin Place - Construction
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Boston, Massachusetts 02215

RFI #0-PB-91: Ground Riser & Bus Bars

Revision	0	**Status**	Closed on 11/24/25
To	Rob Morsi (Suffolk Construction) *(Response Required)*	**From**	Nathan Crowley (Suffolk Construction) 65 Allerton St. Boston, Massachusetts 02119
Date Initiated	Nov 14, 2025	**Due Date**	Nov 21, 2025
Location		**Project Stage**	Bidding
Cost Impact		**Schedule Impact**	
Spec Section		**Cost Code**	
Drawing Number		**Reference**	
Linked Drawings			
Received From		**Sub Job**	
Copies To			

ADDITIONAL FIELDS

Required Reviewers	**Urgency**

Activity

Question

Question from Nathan Crowley Suffolk Construction *on Friday, Nov 14, 2025 at 03:53 PM EST*

Ground bus bars and electrodes are not shown on the plans except for grounding related to the vault, the grade level perimeter ring, and the typical transformer secondary detail. Please provide a grounding riser diagram. Please provide direction for ground bus bars in each electrical room.

Attachments
Grounding RFI.pdf

Official Response

Response from Henry Sniezek Suffolk Construction *on Monday, Nov 24, 2025 at 02:48 PM EST*

Per Payette 11/21 response: A ground bus bar shall be provided in each electric room minimum of 2'-0" x 3" x 1/4". A grounding riser diagram will be provided 1/15 Addendum.

All Replies

Response from Henry Sniezek Suffolk Construction *on Monday, Nov 24, 2025 at 02:48 PM EST*

Per Payette 11/21 response: A ground bus bar shall be provided in each electric room minimum of 2'-0" x 3" x 1/4". A grounding riser diagram will be provided 1/15 Addendum.

GROUNDING GENERAL NOTES

1. ALL GROUNDING CONDUCTOR CONNECTIONS TO CONDUCTORS, GROUND ELECTRODES, AND OTHER BONDING SHALL BE CADWELDED, EXTHERMIC IRREVERSIBLE PROCESS.

2. GROUNDING CONDUCTORS/CONDUITS ENTERING THROUGH FOUNDATION WALLS AND FLOOR SLABS SHALL BE PROVIDE WITH SLEEVES AND WATER STOPS.

3. GROUNDING CONDUCTORS EMBEDED IN TO THE CONCRETE FOUNDATION WORK SHALL BE COORDINATED EARLY IN THE CONSTRUCTION PROCESS TO GET THE APPROPRIATE GROUND CONDUCTOR CONNECTION TO STRUCTURAL STEEL BEFORE CONCRETE IS POURED.

4. GROUND ELECTRODE SYSTEM SHALL BE COORDINATED WITH OTHER UNDERGROUND UTILITIES , ELECTRIC MANHOLES, AND UNDERGROUND ELECTRICAL DUCT BANKS.

5. BOND ANY NEARBY UNDERGROUND METALLIC PIPING SYSTEM, SUCH AS WATER PIPING, GAS SERVICE ETC, TO THE BUILDING UNDERGROUND GROUNDING NETWORK WITH MINIMUM # 1/0 CU.

6. WHERE OTHER GROUNDING NETWORKS ARE NEARBY, SUCH AS LIGHTNING PROTECTION SYSTEM GROUNDS, BOND THESE GROUND NETWORKS TOGETHER FOR EQUIPOTENTIAL WITH A 4/0 CU BARE GROUND CONDUCTOR.

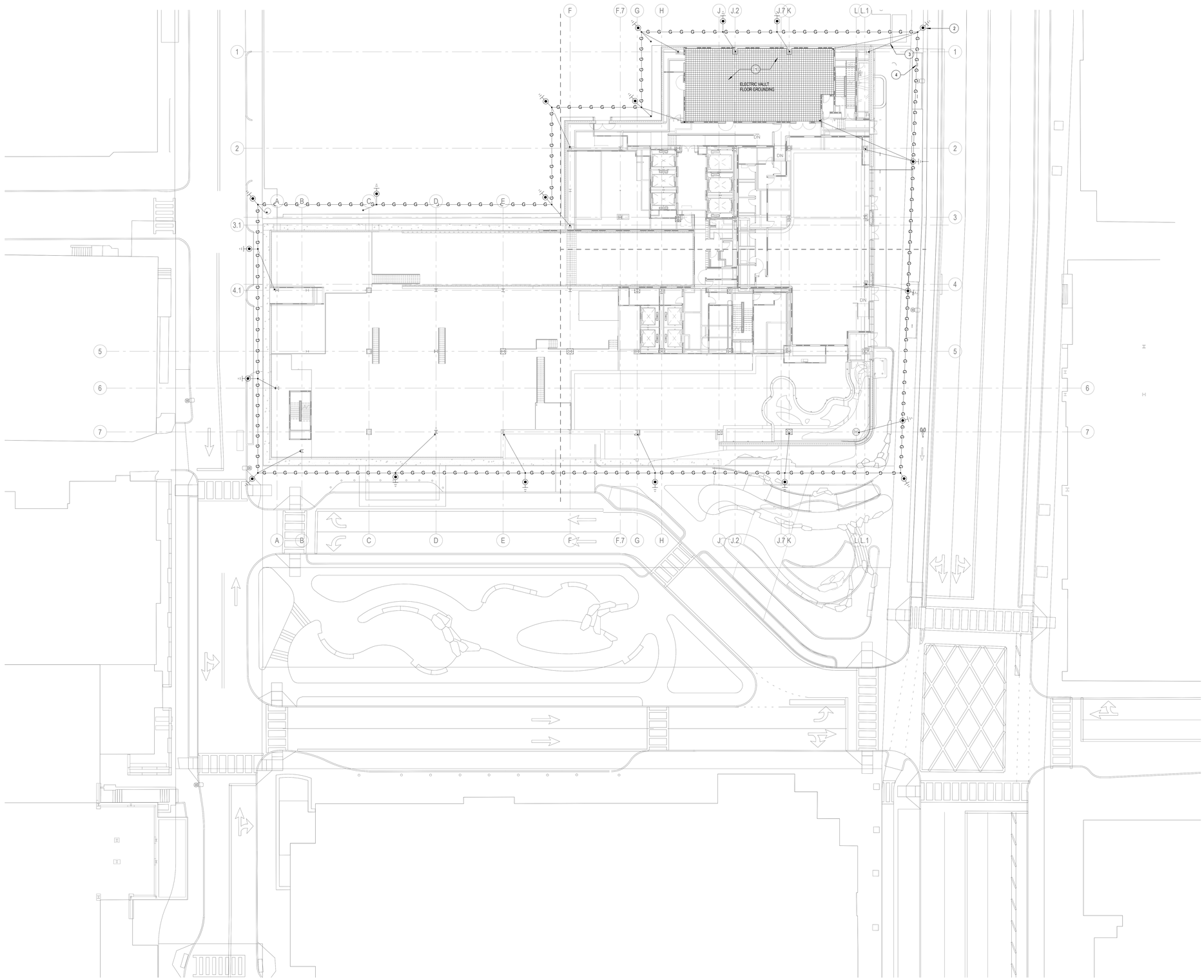
GROUNDING KEY NOTES

1. PROVIDE FLOOR GROUNDING MESH/EQUIPMENT FOR THE ELECTRIC VAULT ROOM FLOOR CONFORM THE UTILITY COMPANY REQUIREMENTS. THE GROUND MESH SHALL BE WIRE GROUNDING MESH WITH 24-INCH SQUARES MADE FROM BARE #6 AWG SOLID COPPER CABLES AS MANUFACTURED BY HARGER CATALOG NUMBER GM1250624CE2 OR SIMILAR .

2. 10'X3/4" GROUNDING RODS TO BE CADWELD CONNECTED TO 500 KCMIL GROUNDING RING. REFER TO SPECIFICATIONS FOR ADDITIONAL INFORMATION. TYPICAL FOR ALL.

3. UNLESS NOTED OTHERWISE ALL THE GROUND WIRING SHOWN ON THIS DRAWING SHALL BE MINIMUM 500KCMIL BARE COPPER.

4. BUILDING PERMETER GROUNDING LOOP. PROVIDE 500 KCMIL COPPER BURIED MIN 30" BELOW GRADE. REFER TO SPECIFICATIONS FOR ADDITIONAL INFORMATION. TYPICAL.



1 GROUND LEVEL - SITE GROUNDING PLAN
1/16" = 1'-0"

Architect

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www.kaizenfood.com

Radiation Shielding Consultant

Nelco/Trapani
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RF Shielding Consultant

PDC Facilities, Inc.
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PAYETTE

Issues

Number	Date	Description
00	10-30-25	FAÇADE AND INFRASTRUCTURE PACKAGE

Revisions

Number	Date	Description

NOT FOR CONSTRUCTION

Dana-Farber Cancer Institute | Beth Israel Lahey Health | Beth Israel Deaconess Medical Center

FUTURE CANCER HOSPITAL

1 JOSLIN PLACE, BOSTON MA 02115

ELECTRICAL GROUNDING SITE PLAN

Drawing Scale
As indicated

Project Number
17000.00

Date Issued
10/30/2025

E0.02

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Autodesk Docs://DFCI-Cancer_Hosp/17000.00-Cancer-Hospital-ELEC-R24.rvt
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LEGEND OF COPPER FEEDER SIZES

NOMINAL AMPERE RATING	FEEDER TAG	CONDUCTORS (3 PHASE, 3 WIRE) WITH GROUND IN CONDUIT	CONDUCTORS (3 PHASE, 4 WIRE) WITH GROUND IN CONDUIT
60	C60Δ	3#4 & 1#10G - 1"C	4#4 & 1#10G - 1 1/4"C
	C60Y		
80	C80Δ	3#3 & 1#8G - 1 1/4"C	4#3 & 1#8G - 1 1/4"C
	C80Y		
100	C100Δ	3#1 & 1#8G - 1 1/2"C	4#1 & 1#8G - 1 1/2"C
	C100Y		
125	C125Δ	3#1/0 & 1#6G - 1 1/2"C	4#1/0 & 1#6G - 2"C
	C125Y		
150	C150Δ	3#1/0 & 1#6G - 1 1/2"C	4#1/0 & 1#6G - 2"C
	C150Y		
175	C175Δ	3#2/0 & 1#6G - 2"C	4#2/0 & 1#6G - 2"C
	C175Y		
200	C200Δ	3#3/0 & 1#6G - 2"C	4#3/0 & 1#6G - 2"C
	C200Y		
225	C225Δ	3#4/0 & 1#4G - 2"C	4#4/0 & 1#4G - 2 1/2"C
	C225Y		
250	C250Δ	3#250 KCMIL & 1#4G - 2 1/2"C	4#250 KCMIL & 1#4G - 2 1/2"C
	C250Y		
300	C300Δ	3#350 KCMIL & 1#4G - 2 1/2"C	4#350 KCMIL & 1#4G - 3"C
	C300Y		
350	C350Δ	3#500 KCMIL & 1#3G - 3"C	4#500 KCMIL & 1#3G - 3 1/2"C
	C350Y		
400	C400Δ	3#600 KCMIL & 1#3G - 3 1/2"C 6#250 KCMIL & 1#3G - 3"C	4#600 KCMIL & 1#3G - 4"C 8#250 KCMIL & 1#3G - 3 1/2"C
	C400Y		
500	C500Δ	2 SETS OF (3#250 KCMIL & 1#2G - 2 1/2"C)	2 SETS OF (4#250 KCMIL & 1#2G - 2 1/2"C)
	C500Y		
600	C600Δ	2 SETS OF (3#350 KCMIL & 1#1G - 2 1/2"C)	2 SETS OF (4#350 KCMIL & 1#1G - 3"C)
	C600Y		
800	C800Δ	2 SETS OF (3#600 KCMIL & 1#1/0G - 3 1/2"C)	2 SETS OF (4#600 KCMIL & 1#1/0G - 4"C)
	C800Y		
1000	C1000Δ	3 SETS OF (3#400 KCMIL & 1#2/0G - 3"C)	3 SETS OF (4#400 KCMIL & 1#2/0G - 3 1/2"C)
	C1000Y		
1200	C1200Δ	3 SETS OF (3#600 KCMIL & 1#3/0G - 3 1/2"C)	3 SETS OF (4#600 KCMIL & 1#3/0G - 4"C)
	C1200Y		
1600	C1600Δ	4 SETS OF (3#600 KCMIL & 1#4/0G - 3 1/2"C)	4 SETS OF (4#600 KCMIL & 1#4/0G - 4"C)
	C1600Y		
2000	C2000Δ	5 SETS OF (3#600 KCMIL & 1#250 KCMIL G - 3 1/2"C)	5 SETS OF (4#600 KCMIL & 1#250 KCMIL G - 4"C)
	C2000Y		
2500	C2500Δ	6 SETS OF (3#600 KCMIL & 1#350 KCMIL G - 3 1/2"C)	6 SETS OF (4#600 KCMIL & 1#350 KCMIL G - 4"C)
	C2500Y		
3000	C3000Δ	8 SETS OF (3#500 KCMIL & 1#400 KCMIL G - 3 1/2"C)	8 SETS OF (4#500 KCMIL & 1#400 KCMIL G - 4"C)
	C3000Y		
3200	C3200Δ	8 SETS OF (3#600 KCMIL & 1#500 KCMIL G - 3 1/2"C)	8 SETS OF (4#600 KCMIL & 1#500 KCMIL G - 4"C)
	C3200Y		
4000	C4000Δ	10 SETS OF (3#600 KCMIL & 1#500 KCMIL G - 3 1/2"C)	10 SETS OF (4#600 KCMIL & 1#500 KCMIL G - 4"C)
	C4000Y		

LEGEND OF FEEDER SIZES NOTES:

1. ALL CONDUCTOR SIZES ARE BASED ON 2023 NEC TABLE 310.16 FOR COPPER CONDUCTORS.
2. CONDUCTORS SHALL BE PROVIDED WITH "MAC" ADAPTERS AS REQUIRED TO COORDINATE WITH BREAKER LUG SIZES.
3. CONDUCTOR TYPE SHALL BE THHN/THWN-2 OR XHHW-2 SUBJECT TO THE RESTRICTIONS OF THE NEC AND PROJECT SPECIFICATIONS.

BR+A DESIGN STANDARDS	DATE 10/25/2024	DRAWING No.

LEGEND OF EMERGENCY SYSTEM FEEDER SIZES

COPPER CONDUCTORS - MINERAL INSULATED (MI)

NOMINAL AMPERE RATING	FEEDER SYMBOL	MI CONDUCTORS (3 PHASE, 3 WIRE)	SUPPLEMENTAL MI GROUND CONDUCTOR SEE NOTE 3	CONDUCTORS IN ENCLOSURE. SEE NOTE 4	MI CONDUCTORS (3 PHASE, 4 WIRE)	SUPPLEMENTAL MI GROUND CONDUCTOR. SEE NOTE 3	CONDUCTORS IN ENCLOSURE. SEE NOTE 4	GROUND CONDUCTOR IN ENCLOSURE. SEE NOTE 5	"S" SEE NOTE 6
60	M60Δ	3#6	N/A	3#2				1#10	3/4"
	M60Y				4#6	N/A	4#2		
80	M80Δ	3#6	N/A	3#2				1#8	3/4"
	M80Y				4#6	N/A	4#2		
100	M100Δ	3#4	N/A	3#1/0				1#8	7/8"
	M100Y				4#4	N/A	4#1/0		
125	M125Δ	3#2	N/A	3#3/0				1#6	1-1/8"
	M125Y				4#2	N/A	4#3/0		
150	M150Δ	3#2	N/A	3#3/0				1#6	1-1/8"
	M150Y				4#2	N/A	4#3/0		
175	M175Δ	3#1	N/A	3#4/0				1#6	1-1/8"
	M175Y				4#1	N/A	4#4/0		
200	M200Δ	3#1/0	N/A	3#4/0				1#6	1-1/8"
	M200Y				4#1/0	N/A	4#4/0		
225	M225Δ	3#1/0	N/A	3#4/0				1#4	1-1/8"
	M225Y				4#1/0	N/A	4#4/0		
250	M250Δ	3#2/0	N/A	3#250 KCMIL				1#4	1-1/4"
	M250Y				4#2/0	N/A	4#250 KCMIL		
300	M300Δ	3#3/0	N/A	3#350 KCMIL				1#4	1-3/8"
	M300Y				4#3/0	N/A	4#350 KCMIL		
350	M350Δ	3#4/0	N/A	3#500 KCMIL				1#3	1-1/2"
	M350Y				4#4/0	N/A	4#500 KCMIL		
400	M400Δ	3#250 KCMIL	N/A	3#600 KCMIL				1#3	1-1/2"
	M400Y				4#250 KCMIL	N/A	4#600 KCMIL		
500	M500Δ	6#2/0	N/A	6#250 KCMIL				1#2	1-1/4"
	M500Y				8#2/0	N/A	8#250 KCMIL		
600	M600Δ	6#3/0	N/A	6#350 KCMIL				1#1	1-3/8"
	M600Y				8#3/0	N/A	8#350 KCMIL		
800	M800Δ	6#250 KCMIL	N/A	6#600 KCMIL				1#1/0	1-1/2"
	M800Y				8#250 KCMIL	N/A	8#600 KCMIL		
1000	M1000Δ	9#4/0	N/A	9#500 KCMIL				1#2/0	1-1/2"
	M1000Y				12#4/0	N/A	12#500 KCMIL		
1200	M1200Δ	9#250 KCMIL	N/A	9#600 KCMIL				1#3/0	1-1/2"
	M1200Y				12#250 KCMIL	N/A	12#600 KCMIL		
1600	M1600Δ	12#250 KCMIL	N/A	12#600 KCMIL				1#4/0	1-1/2"
	M1600Y				16#250 KCMIL	N/A	16#600 KCMIL		
2000	M2000Δ	15#250 KCMIL	5#6	15#600 KCMIL				1#250 KCMIL	1-1/2"
	M2000Y				20#250 KCMIL	N/A	20#600 KCMIL		
2500	M2500Δ	18#350 KCMIL	6#1	18#600 KCMIL				1#350 KCMIL	1-3/4"
	M2500Y				24#350 KCMIL	N/A	24#600 KCMIL		
3000	M3000Δ	24#250 KCMIL	8#4/0	24#500 KCMIL				1#400 KCMIL	1-1/2"
	M3000Y				32#250 KCMIL	8#1/0	32#500 KCMIL		
3200	M3200Δ	24#350 KCMIL	8#250 KCMIL	24#600 KCMIL				1#500 KCMIL	1-3/4"
	M3200Y				32#350 KCMIL	8#2/0	32#600 KCMIL		
4000	M4000Δ	30#250 KCMIL	10#250 KCMIL	30#600 KCMIL				1#500 KCMIL	1-1/2"
	M4000Y				40#250 KCMIL	10#4/0	40#600 KCMIL		

LEGEND OF EMERGENCY SYSTEM FEEDER SIZES NOTES:

1. ALL EMERGENCY SYSTEM FEEDERS SHALL BE UL LISTED AND LABELED, 2 HOUR FIRE RATED MINERAL INSULATED (MI) COPPER POWER CABLES, IN ACCORDANCE WITH NEC ARTICLE 700, AND SHALL BE INSTALLED IN ACCORDANCE WITH MANUFACTURES INSTALLATION INSTRUCTIONS TO MAINTAIN FIRE RATING.
2. ALL CONDUCTOR SIZES ARE BASED ON 2023 NEC TABLE 310.17 AND ARTICLE 332.80(B) FOR CONDUCTORS GROUPED TOGETHER IN A TRIANGULAR OR RECTANGULAR CONFIGURATION, IN FREE AIR.
3. WHERE INDICATED, PROVIDE SUPPLEMENTAL GROUNDING CONDUCTOR TO RUN WITH ASSOCIATED PHASE AND NEUTRAL CONDUCTORS. WHERE SUPPLEMENTAL GROUNDING MI CONDUCTOR PENETRATES PANEL OR OTHER ENCLOSURE, IT SHALL BE PROVIDED WITH A SIZE-TO-SIZE TERMINATION TO TRANSITION TO STRANDED CONDUCTORS. THESE CONDUCTORS SHALL LAND ON THE PANEL GROUND BUS BAR.
4. WHERE MI CONDUCTORS PENETRATE PANEL OR OTHER ENCLOSURE, THEY SHALL BE PROVIDED WITH A SIZED-UP TERMINATION TO TRANSITION TO STRANDED CONDUCTORS, SIZED IN ACCORDANCE WITH 2023 NEC TABLE 310.16. THESE CONDUCTORS SHALL LAND ON THE RESPECTIVE PANEL BUS BARS AND/OR MAIN CIRCUIT BREAKER LUGS. REFER TO SPECIFICATION SECTION 26 0530 FOR ALLOWABLE CONDUCTOR TYPES.
5. FURNISH AND INSTALL A FOOT LUG, ATTACHED TO THE BRASS PLATE HARDWARE INSIDE OF THE PANEL, RUN STRANDED GROUND CONDUCTOR FROM THE LUG TO THE GROUND BUS BAR, SIZED IN ACCORDANCE WITH 2023 NEC 250.122. REFER TO SPECIFICATION SECTION 26 0530 FOR ALLOWABLE CONDUCTOR TYPES.
6. MINIMUM DISTANCE AROUND EXTERIOR OF MI CABLE BUNDLE AND BETWEEN PARALLEL RUNS OF MI CABLE, TO MAINTAIN FREE AIR RATING IN ACCORDANCE WITH 2023 NEC 332.80(B).
7. HORIZONTAL AND VERTICAL SPACING SHALL BE SET TO MAINTAIN MANUFACTURERS UL LISTING REQUIREMENTS FOR THIS APPLICATION.

BR+A DESIGN STANDARDS	DATE 10/25/2024	DRAWING No.

DRY TYPE TRANSFORMER SCHEDULE - COPPER COILS AND FEEDERS

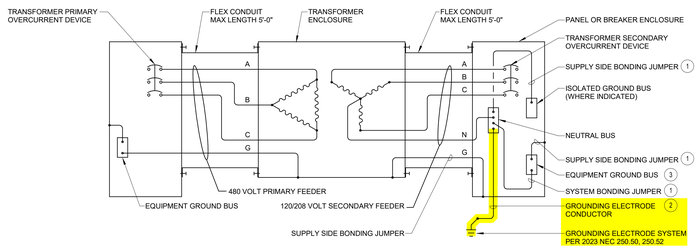
SIZE	kVA	PRIMARY			SECONDARY			SUPPLY SIDE BONDING JUMPER AND SYSTEM BONDING JUMPER (1)	GROUNDING ELECTRODE CONDUCTOR (2)
		AMPERES	480 VOLT OVERCURRENT PROTECTION	CONDUCTORS (3 PHASE, 3 WIRE) WITH GROUND IN CONDUIT	AMPERES	208 VOLT OVERCURRENT PROTECTION	CONDUCTORS (3 PHASE, 4 WIRE) WITH SUPPLY SIDE BONDING JUMPER IN CONDUIT		
CT9	9	11	20A, 3P	3#12 & 1#12G - 3/4"C	25	30A, 3P	4#8 & 1#8G - 1"C	1#8	1#8 - 3/4"C
CT15	15	18	30A, 3P	3#10 & 1#10G - 3/4"C	42	50A, 3P	4#6 & 1#8G - 1"C	1#8	1#8 - 3/4"C
CT30	30	36	60A, 3P	3#4 & 1#10G - 1"C	83	100A, 3P	4#1 & 1#6G - 2"C	1#6	1#6 - 3/4"C
CT45	45	54	80A, 3P	3#3 & 1#8G - 1 1/4"C	125	150A, 3P	4#1/0 & 1#6G - 2"C	1#6	1#6 - 3/4"C
CT75	75	90	150A, 3P	3#1/0 & 1#6G - 1 1/2"C	208	250A, 3P	4#250 KCMIL & 1#2G - 2 1/2"C	1#2	1#2 - 3/4"C
CT112.5	112.5	135	250A, 3P	3#250 KCMIL & 1#4G - 2 1/2"C	313	400A, 3P	4#600 KCMIL & 1#1/0G - 4"C	1#1/0	1#1/0 - 3/4"C
							8#250 KCMIL & 1#1/0G - 3 1/2"C	1#1/0	1#1/0 - 3/4"C
CT150	150	181	300A, 3P	3#350 KCMIL & 1#4G - 2 1/2"C	417	500A, 3P	2 SETS OF (4#250 KCMIL & 1#1/0G - 2 1/2"C)	1#1/0	1#1/0 - 3/4"C
CT225	225	271	400A, 3P	3#600 KCMIL & 1#3G - 3 1/2"C	625	800A, 3P	2 SETS OF (4#600 KCMIL & 1#3/0G - 4"C)	1#3/0	1#3/0 - 1"C
				6#250 KCMIL & 1#3G - 3"C					
CT300	300	361	600A, 3P	2 SETS OF (3#350 KCMIL & 1#1G - 2 1/2"C)	834	1000A, 3P	3 SETS OF (4#400 KCMIL & 1#3/0G - 3 1/2"C)	1#3/0	1#3/0 - 1"C
CT500	500	602	1000A, 3P	3 SETS OF (3#400 KCMIL & 1#2/0G - 3"C)	1390	1600A, 3P	4 SETS OF (4#600 KCMIL & 1#300 KCMIL G - 4"C)	1#300 KCMIL	1#3/0 - 1"C

DRY TYPE TRANSFORMER SCHEDULE NOTES:

1. ALL PHASE AND NEUTRAL CONDUCTOR SIZES ARE BASED ON COPPER CONDUCTORS PER 2023 NEC TABLE 310.16.
2. MANUFACTURERS SELECTION FOR THE TYPE OF PRIMARY CIRCUIT BREAKER (THERMAL MAGNETIC VS. SOLID STATE) SHALL ENSURE COORDINATION WITH TRANSFORMER IN-RUSH CURRENT. IF MIS-COORDINATION IS IDENTIFIED BY THE COORDINATION STUDY, THE MANUFACTURER SHALL REPLACE THE DEVICE WITH A DEVICE THAT WILL PROPERLY COORDINATE, AT NO ADDITIONAL COST.
3. SECONDARY OVERCURRENT PROTECTION SHALL BE LOCATED WITHIN A PANELBOARD (MAIN CIRCUIT BREAKER) OR INDIVIDUALLY MOUNTED CIRCUIT BREAKER. THE SECONDARY OVERCURRENT PROTECTION DEVICE SHALL BE LOCATED SUCH THAT THE MAXIMUM LENGTH OF SECONDARY CONDUCTORS DO NOT EXCEED 10'-0".
4. CONDUCTORS SHALL BE PROVIDED WITH "MAC" ADAPTERS AS REQUIRED TO COORDINATE WITH BREAKER LUG SIZES.
5. CONDUCTOR TYPE SHALL BE THHN/THWN-2 OR XHHW-2 SUBJECT TO THE RESTRICTIONS OF THE NEC AND PROJECT SPECIFICATIONS.

DRY TYPE TRANSFORMER KEY NOTES:

1. SIZE OF SUPPLY SIDE BONDING JUMPER AND SYSTEM BOUNDING JUMPER, PER 2023 NEC 250.102(C)(1).
2. SIZE OF GROUNDING ELECTRODE CONDUCTOR. THE GROUNDING ELECTRODE CONDUCTOR SHALL BE UNSPLICED AND RUN IN CONDUIT FROM THE NEUTRAL BUS TO THE GROUDING ELECTRODE SYSTEM, PER 2023 NEC 250.66.
3. SECONDARY EQUIPMENT GROUND BUS SHALL BE LOCATED WITHIN A PANELBOARD OR INDIVIDUALLY MOUNTED CIRCUIT BREAKER.



DRY TYPE TRANSFORMER WIRING DIAGRAM

BR+A DESIGN STANDARDS	DATE 05/06/2025	DRAWING No.

SPECIAL PURPOSE RECEPTACLE SCHEDULE

SYMBOL	NEMA	DESCRIPTION	HUBBELL CAT. No.	COOPER CAT. No.	BRYANT CAT. No.	P & S CAT. No.	C.B.	BRANCH CIRCUIT SEE NOTES
1	5-20R	20A - 125V, 2P, 3W	HBL5361	5361	5361	5361	20A, 1P	3#10 & 1#10G-3/4"C
2	5-30R	30A - 125V, 2P, 3W	HBL9308	5716-N	9530-FR	3802	30A, 1P	3#10 & 1#10G-3/4"C
3	5-50R	50A - 125V, 2P, 3W	HBL9360	1253	9550-FR	---	50A, 1P	2#6 & 1#10G-3/4"C
4	6-20R	20A - 250V, 2P, 3W	HBL5461	5461	5461	5871	20A, 2P	3#10 & 1#10G-3/4"C
5	6-30R	30A - 250V, 2P, 3W	HBL9330	5700-N	9630-FR	3801	30A, 2P	3#10 & 1#10G-3/4"C
6	6-50R	50A - 250V, 2P, 3W	HBL9367	5709-N	9650-FR	3804	50A, 2P	3#6 & 1#10G-3/4"C
7	14-20R	20A - 125/250V, 3P, 4W	HBL8410	5759	---	3820	20A, 2P	3#10 & 1#10G-3/4"C
8	14-30R	30A - 125/250V, 3P, 4W	HBL9430A	5744-N	9430-FR	3864	30A, 2P	3#10 & 1#10G-3/4"C
9	14-50R	50A - 125/250V, 3P, 4W	HBL9450A	5754-N	9450-FR	3894	50A, 2P	3#6 & 1#10G-1"C
10	14-60R	60A - 125/250V, 3P, 4W	HBL9460A	9460-N	9460-FR	---	60A, 2P	3#6 & 1#10G-1"C
11	15-20R	20A - 250V, 3Ø, 3P, 4W	HBL8420A	---	---	---	20A, 3P	3#12 & 1#12G-3/4"C
12	15-30R	30A - 250V, 3Ø, 3P, 4W	HBL8430A	8430-N	8430-FR	5740	30A, 3P	3#10 & 1#10G-3/4"C
13	15-50R	50A - 250V, 3Ø, 3P, 4W	HBL8450A	8450-N	8450-FR	5750	50A, 3P	3#6 & 1#10G-1"C
14	15-60R	60A - 250V, 3Ø, 3P, 4W	HBL8460A	8460-N	8460-FR	5760	60A, 3P	3#6 & 1#10G-1"C
15	L5-20R	20A - 125V, 2P, 3W, TWIST LOCK	HBL2310	CWL520R	70520-FR	L520R	20A, 1P	3#10 & 1#10G-3/4"C
16	L5-30R	30A - 125V, 2P, 3W, TWIST LOCK	HBL2610	CWL530R	70530-FR	L530R	30A, 1P	3#10 & 1#10G-3/4"C
17	L6-20R	20A - 250V, 2P, 3W, TWIST LOCK	HBL2320	CWL620R	70620-FR	L620R	20A, 2P	3#10 & 1#10G-3/4"C
18	L6-30R	30A - 250V, 2P, 3W, TWIST LOCK	HBL2620	CWL630R	70630-FR	L630R	30A, 2P	3#10 & 1#10G-3/4"C
19	L14-20R	20A - 125/250V, 3P, 4W, TWIST LOCK	---	---	---	L1420R	20A, 2P	3#10 & 1#10G-3/4"C
20	L14-30R	30A - 125/250V, 3P, 4W, TWIST LOCK	---	---	---	L1430R	30A, 2P	3#10 & 1#10G-3/4"C
21	L15-20R	20A - 250V, 3Ø, 3P, 4W, TWIST LOCK	---	---	---	L1520R	20A, 3P	3#12 & 1#12G-3/4"C
22	L15-30R	30A - 250V, 3Ø, 3P, 4W, TWIST LOCK	---	---	---	L1530R	30A, 3P	3#10 & 1#10G-3/4"C
23	L21-20R	20A - 125/250V, 3Ø, 4P, 5W, TWIST LOCK	HBL2510	---	---	L2120R	20A, 3P	4#12 & 1#12G-3/4"C

NOTES:

1. BRANCH CIRCUIT FOR DEVICE SYMBOLS 1, 2, 15 & 16 INCLUDES A SPARE PHASE CONDUCTOR FOR FUTURE USE. WITHIN THE PANEL, SPARE PHASE CONDUCTOR SHALL BE CAPPED, TY-WRAPPED TO ASSOCIATED PHASE, NEUTRAL AND GROUND CONDUCTOR, PROPERLY IDENTIFIED AND INCLUDE SUFFICIENT SLACK TO CONNECT TO CIRCUIT BREAKER IN THE FUTURE. WITHIN THE OUTLET BOX, SPARE PHASE CONDUCTOR SHALL BE CAPPED, PROPERLY IDENTIFIED AND INCLUDE SUFFICIENT SLACK TO CONNECT TO WIRING DEVICE IN THE FUTURE.
2. BRANCH CIRCUIT FOR DEVICE SYMBOLS 4, 5, 17 & 18 INCLUDES A SPARE NEUTRAL CONDUCTOR FOR FUTURE USE. WITHIN THE PANEL, SPARE NEUTRAL CONDUCTOR SHALL TERMINATE ON THE NEUTRAL BUS AND BE PROPERLY IDENTIFIED. WITHIN THE OUTLET BOX, SPARE NEUTRAL CONDUCTOR SHALL BE CAPPED, PROPERLY IDENTIFIED AND INCLUDE SUFFICIENT SLACK TO CONNECT TO WIRING DEVICE IN THE FUTURE.

MOTOR WIRING SCHEDULE

(FOR SINGLE SPEED, 480 VOLT MOTORS ONLY)

HORSE POWER	FEEDER - 480VOLT, 3Ø, 3WIRE
1/2	3#12 & 1#12G - 3/4"C.
3/4	3#12 & 1#12G - 3/4"C.
1	3#12 & 1#12G - 3/4"C.
1 1/2	3#12 & 1#12G - 3/4"C.
2	3#12 & 1#12G - 3/4"C.
3	3#12 & 1#12G - 3/4"C.
5	3#10 & 1#10G - 3/4"C.
7.5	3#10 & 1#10G - 3/4"C.
10	3#8 & 1#10G - 1"C.
15	3#8 & 1#8G - 1"C.
20	3#6 & 1#8G - 1"C.
25	3#6 & 1#8G - 1"C.
30	3#4 & 1#6G - 1 1/4"C.
40	3#3 & 1#6G - 1 1/4"C.
50	3#1 & 1#4G - 1 1/2"C.
60	3#1/0 & 1#4G - 2"C.
75	3#2/0 & 1#4G - 2"C.
100	3#3/0 & 1#3G - 2"C.
125	3#4/0 & 1#2G - 2 1/2"C.
150	3#350KCMIL & 1#1/0G - 3"C.
200	3#500KCMIL & 1#2/0G - 3 1/2"C.

MOTOR WIRING SCHEDULE

(FOR SINGLE SPEED, 208 VOLT MOTORS ONLY)

HORSE POWER	FEEDER - 208VOLT, 3Ø, 3WIRE
1/2	3#12 & 1#12G - 3/4"C.
3/4	3#12 & 1#12G - 3/4"C.
1	3#12 & 1#12G - 3/4"C.
1 1/2	3#12 & 1#12G - 3/4"C.
2	3#10 & 1#10G - 3/4"C.
3	3#10 & 1#10G - 3/4"C.
5	3#8 & 1#8G - 1"C.
7.5	3#6 & 1#8G - 1"C.
10	3#4 & 1#6G - 1 1/4"C.
15	3#3 & 1#6G - 1 1/4"C.
20	3#2 & 1#6G - 1 1/4"C.
25	3#1 & 1#4G - 1 1/2"C.
30	3#1/0 & 1#4G - 2"C.
40	3#3/0 & 1#3G - 2"C.
50	3#4/0 & 1#2G - 2 1/2"C.
60	3#300 KCMIL & 1#1G - 3"C.
75	3#500 KCMIL & 1#1/0G - 3 1/2"C.
100	3#600 KCMIL & 1#2/0G - 3 1/2"C.

BRANCH CIRCUITS SCHEDULE

CIRCUIT BREAKER	CONDUCTOR
120 or 277 VOLT 1Ø, 2W. CIRCUITS	
30A-1P	2#10 & 1#10G - 3/4"C
40A-1P	2#8 & 1#10G - 3/4"C
50A-1P	2#6 & 1#10G - 3/4"C
60A-1P	2#4 & 1#10G - 1"C
208 VOLT 1Ø, 2W. CIRCUITS	
20A-2P	2#12 & 1#12G - 3/4"C
30A-2P	2#10 & 1#10G - 3/4"C
40A-2P	2#8 & 1#10G - 3/4"C
50A-2P	2#6 & 1#10G - 3/4"C
60A-2P	2#4 & 1#10G - 1"C
120/208 VOLT 1Ø, 3W. CIRCUITS	
20A-3P	3#12 & 1#12G - 3/4"C
30A-3P	3#10 & 1#10G - 3/4"C
40A-3P	3#8 & 1#10G - 3/4"C
50A-3P	3#6 & 1#10G - 3/4"C
60A-3P	3#4 & 1#10G - 1"C
208 or 480 VOLT 3Ø, 3W. CIRCUITS	
20A-3P	3#12 & 1#12G - 3/4"C
30A-3P	3#10 & 1#10G - 3/4"C
40A-3P	3#8 & 1#10G - 3/4"C
50A-3P	3#6 & 1#10G - 3/4"C
60A-3P	3#4 & 1#10G - 1"C
208 or 480 VOLT 3Ø, 4W. CIRCUITS	
20A-3P	4#12 & 1#12G - 3/4"C
30A-3P	4#10 & 1#10G - 3/4"C
40A-3P	4#8 & 1#10G - 3/4"C
50A-3P	4#6 & 1#10G - 3/4"C
60A-3P	4#4 & 1#10G - 1 1/4"C

NOTES:
1. FOR BRANCH CIRCUITS INDICATED ON PLAN

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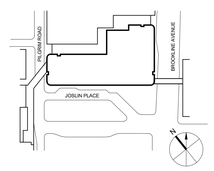
Issues

Number	Date	Description
.00	10-30-25	FAÇADE AND INFRASTRUCTURE PACKAGE

Revisions

Number	Date	Description

NOT FOR CONSTRUCTION



Dana-Farber Cancer Institute | Beth Israel Lahey Health | Beth Israel Deaconess Medical Center

FUTURE CANCER HOSPITAL
1 JOSLIN PLACE, BOSTON MA 02115

ELECTRICAL GENERAL SCHEDULE

Drawing Scale: 12" = 1'-0"
Project Number: 17000.00
Date Issued: 10/30/2025

E7.01

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6. Performance Testing for Hospital Grade devices installed within Patient Care Spaces:

    a. Hospital Grade devices installed in Category 1 and Category 2 Patient Care Areas, including, but not limited to; patient bed locations and locations where deep sedation and general anesthesia are administered, shall be performance tested as follows:

        1) The physical integrity of each receptacle shall be confirmed by visual inspection.
        2) The continuity of the grounding circuit in each electrical receptacle shall be verified.
        3) Correct polarity of the hot and neutral connections in each electrical receptacle shall be confirmed.
        4) The retention force of the grounding blade of each electrical receptacle (except locking-type receptacles) shall be not less than 115 g (4 oz).”

    b. Test instruments shall comply with UL 1436.
    c. Devices that fail the above performance testing shall be replaced, and the new device tested, until all spaces pass the performance testing.
    d. Documentation shall be provided, in the close-out submittal, indicating every room has passed this performance test.

M. Grounding

1. Install a complete equipment grounding system. Equipment grounding system shall be designed so metallic structures, enclosures, raceways, junction boxes, outlet boxes, cabinets, machine frames, metal fences, portable equipment and other conductive items in proximity to electrical circuits operate continuously at ground potential and provide low impedance path for possible ground fault currents.
2. System shall meet NEC requirements, and as shown on the drawings and as specified. Bonding jumpers shall be installed at all locations required by NEC.
3. A main electrical service copper ground bus shall be installed surrounding the electric room. The bus shall be 3" by 1/4" in cross-section and supported off the walls with 3/4" spacers on 2'-0" centers, mounted a maximum of 36” AFF except where ground bus offset over doors and other obstructions. All electrical equipment in the electric room shall be properly bonded to the ground bus with an adequately sized copper conductor as well as the driven ground rod system external to the building.
4. Provide a complete grounding network, per NEC requirements, for main service equipment, consisting of, but not limited to, one or more of the following:

    a. Ground loop buried around the building, 500 kcMIL bare copper, with down conductors from main electrical service ground bus in the main electric room. Down conductors shall be 500 kcMIL copper conductors in conduit to the ground loop.
    b. Appropriate number of 3/4" diameter, 10'-0" long copper-clad steel ground rod clusters and copper ground wire connections from grounding loop.
    c. Connections from ground loop to the building structural steel. Where the steel structure is discontinuous, each section of structure shall be connected to the grounding network. Connections to building steel shall be 500 kcMIL copper conductors to the ground loop.
    d. Metallic underground water pipe in direct contact with earth for a minimum of 10’-0”, bonded to the main service ground bus. Ground conductors to metallic piping shall be 500 kcMIL copper conductors in conduit

e. Metal in-ground support structure(s) in direct contact with earth vertically for a minimum of 10’-0”.
f. An electrode, consisting of one or more bare, zinc galvanized or other electrically conductive coated steel reinforcing bars or rods, a minimum of 20’ long and 0.5” diameter; or a bare copper conductor, a minimum of 20’ long and #4AWG. Electrode or bare copper conductor shall be encased in a minimum of 2” concrete and located horizontally within the building foundation or footing that is in direct contact with earth.
g. Electrolytic Ground Electrodes

1) Electrolytic Ground Electrodes and Backfill Materials shall be utilized where soil conditions prohibit optimum performance of installed grounding system.
2) Electrolytic ground rod shall be UL 467 listed, made with Type K copper 0.083” wall thickness and 2.125” OD. They shall be filled with non-hazardous electrolytic salts, and supplied with adequate amounts of self-compacting backfill material. Kits shall be supplied with Poly-plastic test wells with tensile strength of 3100-5500PSI, and impact strength of 5-15Ft.lbs/in.
3) The vertical model of electrolytic ground electrode shall provide the copper conductor exothermic connection point 914 mm (3 feet) below grade. The horizontal model of electrolytic ground rod shall provide the copper conductor exothermic connection point on horizontal shaft 914 mm (3 feet) below grade. The copper ground connection shall provide continuous loop to ground conductor counterpoise to limit inductive impedance for dissipation of lightning or system fault current.
4) The electrolytic ground rod shall have minimum thirty (30) year replacement warranty.
5) Backfill shall be carbon based, hygroscopic, noncorrosive, electrically conductive (<0.35 ohm-cm), pH (8.7), and shall be RoHS compliant. No mixing or grouting pump shall require for installation. The backfill material shall be self-compacting.
6) Electrolytic ground electrodes shall be manufactured by nVent Erico, Lyncole or equal.

5. Ground points shall be driven to achieve 5 ohms maximum resistance to ground. Provide additional grounding as required to achieve 5 ohms overall. Where geological conditions dictate, ground wire mesh may be provided or additional rods shall be driven in compacted earth areas as required to meet 5 ohm resistance requirement. Plate electrodes may be used where ground rods cannot be driven to achieve grounding requirements.
6. All grounding connections, including to ground loop system, building steel, etc. shall be made with an exothermic welding process. Ground connections to metallic piping shall be with approved grounding clamp.
7. The building grounding grid system shall be as follows:

a. 3/4" by 10'-0" driven ground rods shall be composed of copper clad steel. Tops of all buried ground rods shall be a minimum of 6" below grade. Where rock is encountered, grounding plates may be used in lieu of grounding rods. Grounding plates shall be copper. Sizes of plates shall be as required.
b. All grounding connections, including to ground loop system, building steel, etc. shall be made with an Exothermic process. Exothermic welded grounding connections shall be manufactured by nVent Erico – Cadweld, ABB Thomas & Betts, Furse Weld or approved equal.

c. Exposed grounding connections and ground connections to metallic piping shall be with approved grounding clamp. Exposed connections between different metals shall be sealed with no-oxide paint Grade A.
d. Dry season resistance of the grounding system shall not exceed 5 ohms. If such resistance cannot be obtained with the grounding system indicated on the drawings, the Electrical Subcontractor shall provide additional grounding as required and as directed by the Architect, to provide the 5 ohms resistance value.
e. Testing per the requirements of IEEE 81 shall be conducted at the completion of the grounding system installation in the presence of the Architect or their designated representative. The Electrical Subcontractor shall pay for all materials and workmanship. Defects appearing under the tests shall be corrected by the Electrical Subcontractor.
f. An interior and exterior buried grounding grid system shall connect the building steel and lightning protection system to driven ground rods.

8. Provide separate green insulated equipment grounding conductor for each <u>single- or 3-phase feeder</u>. Install grounding conductor in common conduit with related phase and neutral conductors. Where parallel feeders are installed in more than (1) raceway, an individual, full size green insulated equipment ground conductor shall be provided in each.
9. Provide separate green insulated equipment grounding conductor for each <u>branch circuit</u>. Where multiple branch circuits (3 maximum) are installed in a common raceway or prefabricated conductor assembly (i.e. MC or AC cable), a minimum of one ground conductor shall be included, provided that each of the branch circuits originate from a different phase. Install grounding conductor in common conduit or prefabricated conductor assembly with related phase and neutral conductors.
10. Determine numbers and sizes of screw terminals for equipment grounding bars in panelboards and other electrical equipment. Provide screw terminals for active circuits, spares and spaces.
11. Provide green, insulated equipment ground conductor in same raceway with associated phase conductors, as follows:

    a. From main service ground to ground bus in service entrance equipment to ground bus in all distribution panels, remote panels, motor control centers, etc., size as shown on the drawings.
    b. From green ground terminals of receptacles to green 10-32 washer-in-head outlet box machine screw and to panelboard grounding bus. (Receptacles with special cast boxes and factory designed and approved ground path do not require separate ground jumper.)
    c. From panelboard ground bus to green 10-32 washer-in-head machine screw in ceiling outlet box or junction box, through flexible metallic conduit to ground terminal on lighting fixture and from green 10-32 washer-in-head machine screw in ceiling outlet box or junction box through flexible metallic conduit to green 10-32 washer-in-head machine screw in switch outlet box.
    d. From panelboard ground bus to green 10-32 washer-in-head machine screw in junction box or disconnect switch through flexible metallic conduit to ground terminal in connection box mounted on single-phase fractional horsepower motor.
    e. From equipment ground bus in motor control center through conduit and flexible metallic conduit to ground terminal in connection box mounted on 3-phase motors. Ground conductors for motors with separate starters and disconnect devices shall originate at ground bar in panelboard and shall be bonded to each starter and disconnect device enclosure.
    f. From dry type transformer neutrals to the building steel (or the ground grid system) by means of copper wire, as scheduled on the drawings.

12. Provide green insulated grounding conductor in all non-metallic conduits.
13. Normal and Critical Branch Panelboard Bonding:

    a. In accordance with National Electrical Code Article 517.14, the equipment grounding terminal buses of the following branch circuit panelboards shall be connected together with a continuous, green insulted, copper conductor, not smaller than #10 AWG:

        1) Normal and Critical Branch panelboards, serving the same, individual patient care vicinity.
        2) Critical panelboards served from separate transfer switches, serving the same, individual patient care vicinity.

    b. The bonding conductor shall be permitted to be broken, in order to terminate it on the equipment grounding terminal bus in each panelboard.

14. All empty conduit runs shall be provided with insulated and grounding bushing and grounded by a #12 AWG green ground conductor to the nearest panel ground bus.

3.3 ADDITIONAL INSTALLATION REQUIREMENTS FOR SPECIAL SYSTEMS

A. Emergency Power Supply and Distribution Systems

1. The following shall be installed within 2 hour fire rated rooms, closets and/or shafts or shall be 2 hour fire rated type MI (Mineral-Insulated metal-sheathed) cable, whether indicated on the drawings or not.
2. Where installed above finished ceilings, the fire suppression system shall be also be installed above the ceiling to protect the area**.**

    a. Feeders from the generator(s) to the main stand-by distribution equipment.
    b. Feeders from main stand-by distribution equipment to each life safety automatic transfer switch.
    c. Feeder from main stand-by distribution equipment to the docking station used for temporary roll up generator or load bank.
    d. Feeders from each life safety automatic transfer switch to its associated distribution equipment.
    e. Feeders from the life safety distribution equipment to all downstream switchboards, panelboards, transformers, enclosed circuit breakers, etc.
    f. Branch circuits from associated distribution equipment to the following systems, including all equipment associated with each system:

        1) Smoke evacuation/exhaust systems
        2) Stairwell pressurization systems
        3) Elevator pressurization systems
        4) Atrium exhaust systems
        5) Elevator cab lighting

    g. Engine start circuit from each life safety automatic transfer switch to the emergency generator.
    h. Control wiring associated with the following systems:

        1) Smoke evacuation/exhaust systems
        2) Stairwell pressurization systems